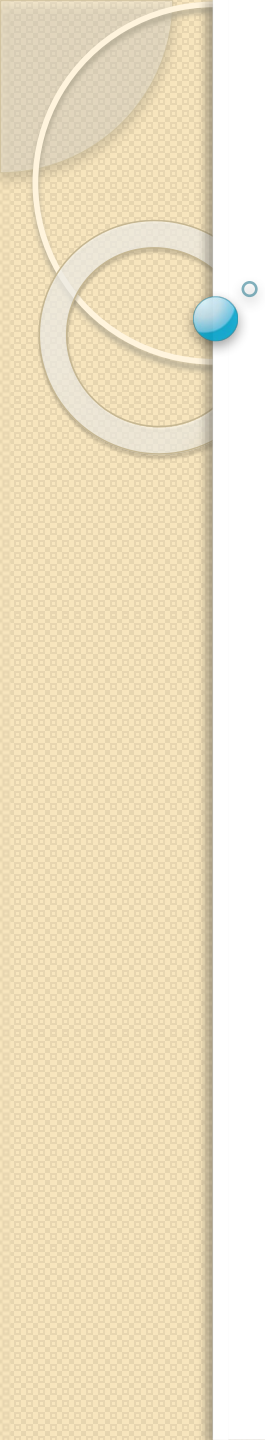


CHAPTER FIVE

Link Layer and Physical Layer



Data Link Layer

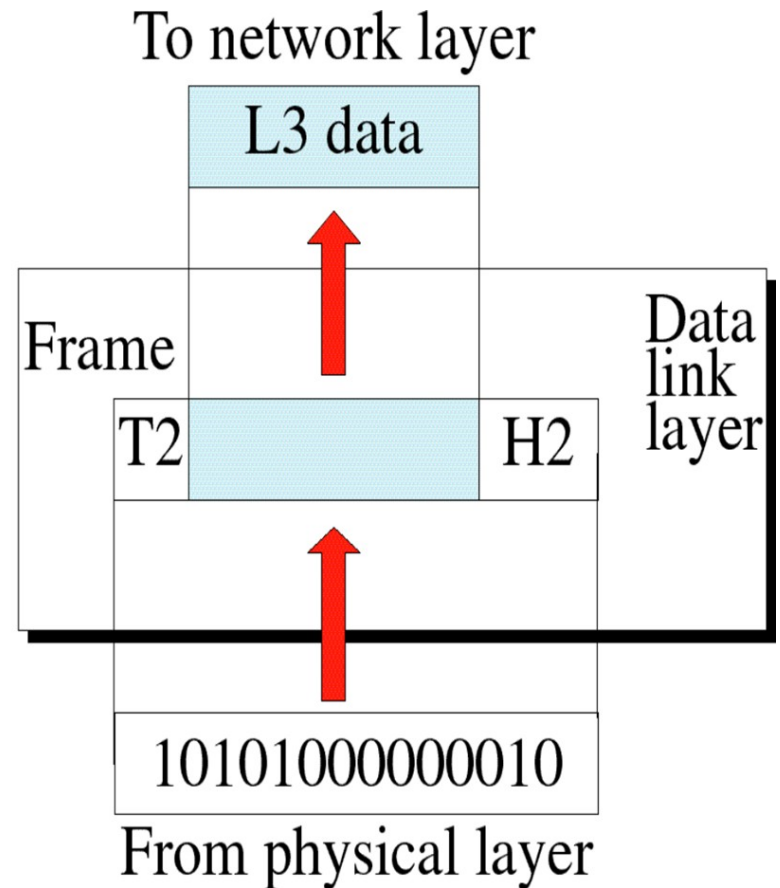
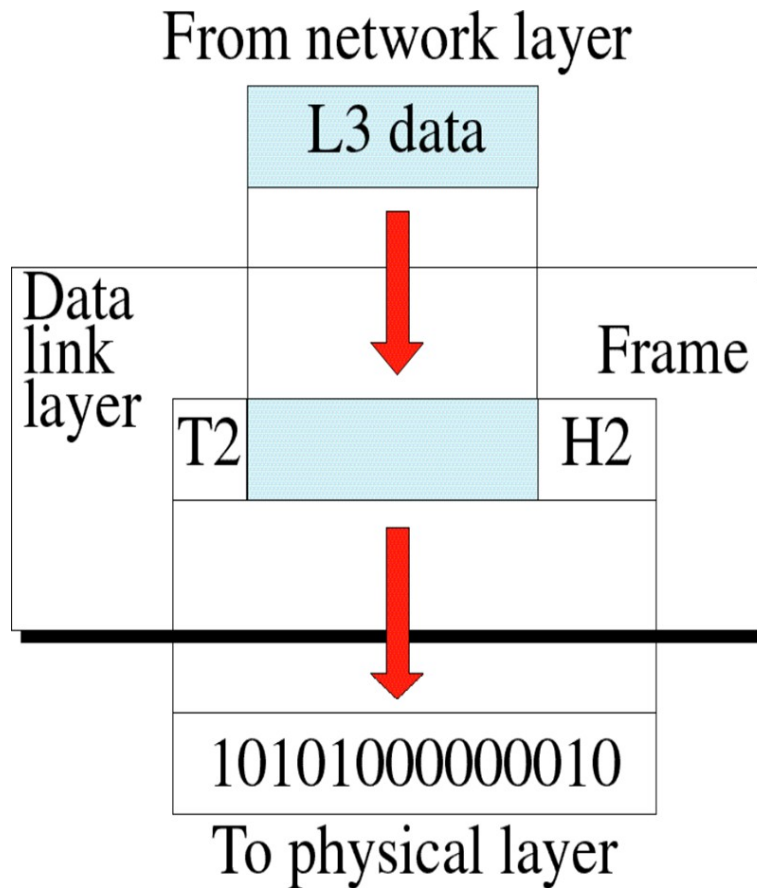
Data Link Layer

- The data link layer is responsible for moving frames from one hop (node) to the next.
- The Data Link layer is responsible for communications between end-device network interface cards.

Data Link Layer

- It allows upper layer protocols to access the physical layer media and encapsulates Layer 3 packets (IPv4 and IPv6) into Layer 2 Frames.
- The data link layer receives the packets from network layer and creates FRAMES , add physical address to these frames & pass them to physical layer
- It also performs error detection and rejects corrupted frames.

Cont..



Link layer services

- **Framing** : data link layer adds header and trailer to the data received from network layer and turns it into frames.
(Frame contains all the addressing information necessary to travel from Source to Destination).
- **Flow Control**: data link layer prevents the fast sender from drowning the slow receiver.

Link layer services

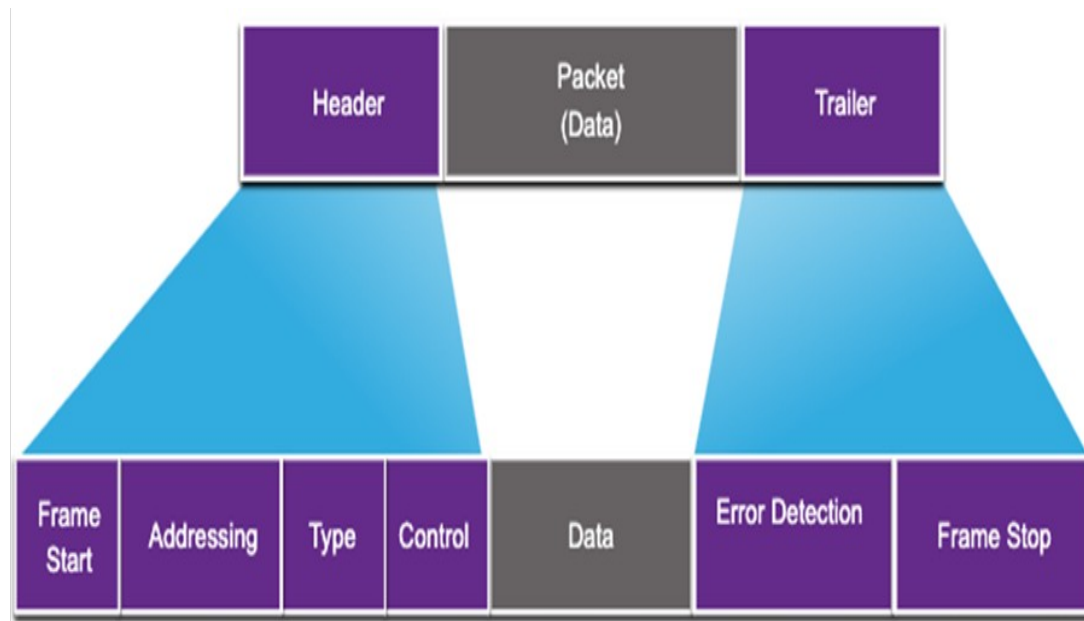
- **Error Control:** It provides the mechanism of error control in which it detects & retransmits damaged or lost frames.
- **Access Control:** When single communication Channel is shared by multiple devices, MAC layer of data link layer provides help to determine which device has control over the channel.

Framing

- Data is encapsulated by the data link layer with a header and a trailer to form a frame.
- A data link frame has three parts:
 - Header
 - Data
 - Trailer

The fields of the header and trailer vary according to data link layer protocol.

The amount of control information carried with in the frame varies according to access control information and logical topology.



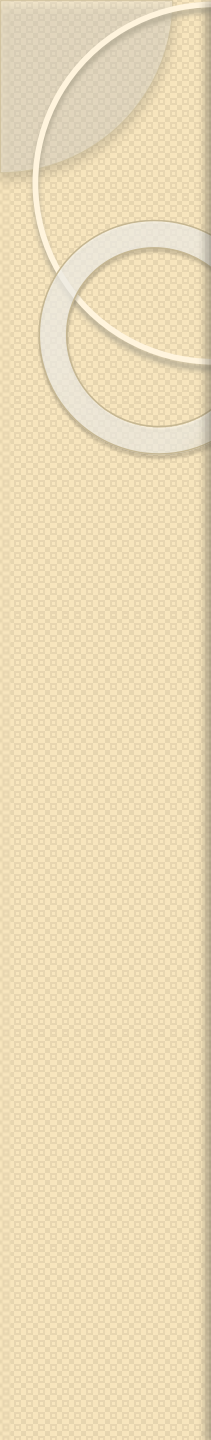
Field	Description
Frame Start and Stop	Identifies beginning and end of frame
Addressing	Indicates source and destination nodes
Type	Identifies encapsulated Layer 3 protocol
Control	Identifies flow control services
Data	Contains the frame payload
Error Detection	Used for determine transmission errors

Data Link Sub Layers

- The Data Link Layer consists of two sublayers. **Logical Link Control (LLC)** and **Media Access Control (MAC)**.
 - The **LLC sublayer** communicates between the networking software at the upper layers and the device hardware at the lower layers.
 - Places information in the frame to identify which network layer protocol is used for the frame.
 - it controls the framing, flow control, and error checking functions of the data link layer.

Data Link Sub Layers

- The **MAC sublayer** is responsible for data encapsulation and media access control.
 - Responsible for data encapsulation and media access control, and provides data link layer addressing.
 - It provides control for accessing the transmission medium.



Network	Network Layer Protocol			
Data Link	LLC Sublayer	LLC Sublayer - IEEE 802.2		
	MAC Sublayer	Ethernet IEEE 802.3	WLAN IEEE 802.11	WPAN IEEE 802.15
Physical		Various Ethernet standards for Fast Ethernet, Gigabit Ethernet, etc.	Various WLAN standards for different types of wireless communications	Various WPAN standards for Bluetooth, RFID, etc.

Multiple Access Protocols

- These protocols allow a number of nodes or users to access a shared network channel.
- Multiple Access Protocols required to decrease collision.

Eg. All nodes operating in half-duplex, competing for use of the medium.

Multiple Access Protocols

- Carrier sense multiple access with collision detection (CSMA/CD) as used on legacy bus-topology Ethernet.
- Carrier sense multiple access with collision avoidance (CSMA/CA) as used on Wireless LANs.

CSMA/CD and CSMA/CA

CSMA/CD

- Used by Ethernet LANs.
- Operates in half-duplex mode where only one device sends or receives at a time.
- Uses a collision detection process to govern when a device can send and what happens if multiple devices send at the same time.

CSMA/CD and CSMA/CA

CSMA/CD collision detection process:

- Devices transmitting simultaneously will result in a signal collision on the shared media.
- Devices detect the collision.
- Devices wait a random period of time and retransmit data.

Cont..

- CSMA/CA

- Used by IEEE 802.11 WLANs.
- Operates in half-duplex mode where only one device sends or receives at a time.
- Uses a collision avoidance process to govern when a device can send and what happens if multiple devices send at the same time.

Cont..

CSMA/CA collision avoidance process:

- When transmitting, devices also include the time duration needed for the transmission.
- Other devices on the shared medium receive the time duration information and know how long the medium will be unavailable.

Data Link and its responsibilities

Data Link Control

- When a data frame (Layer-2 data) is sent from one host to another over a single medium, it is required that the sender and receiver should work at the same speed.
- That is, sender sends at a speed on which the receiver can process and accept the data.
- What if the speed (hardware/software) of the sender or receiver differs?
- If sender is sending too fast the receiver may be overloaded, (swamped) and data may be lost.

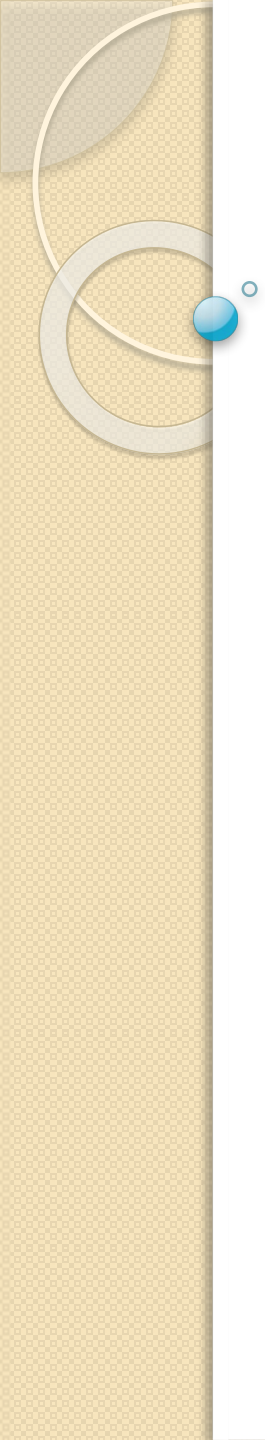
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Error detection and correction

- Error control is both error detection and error correction.
- It allows the receiver to inform the sender of any frames lost or damaged in transmission and coordinates the retransmission of those frames by the sender.
- In the data link layer, the term error control refers primarily to methods of error detection and retransmission.

Cont..

- Error control in the data link layer is often implemented simply: Any time an error is detected in an exchange, specified frames are retransmitted.
- This process is called **automatic repeat request (ARQ)**.
- Error control in the data link layer is based on automatic repeat request, which is the retransmission of data.



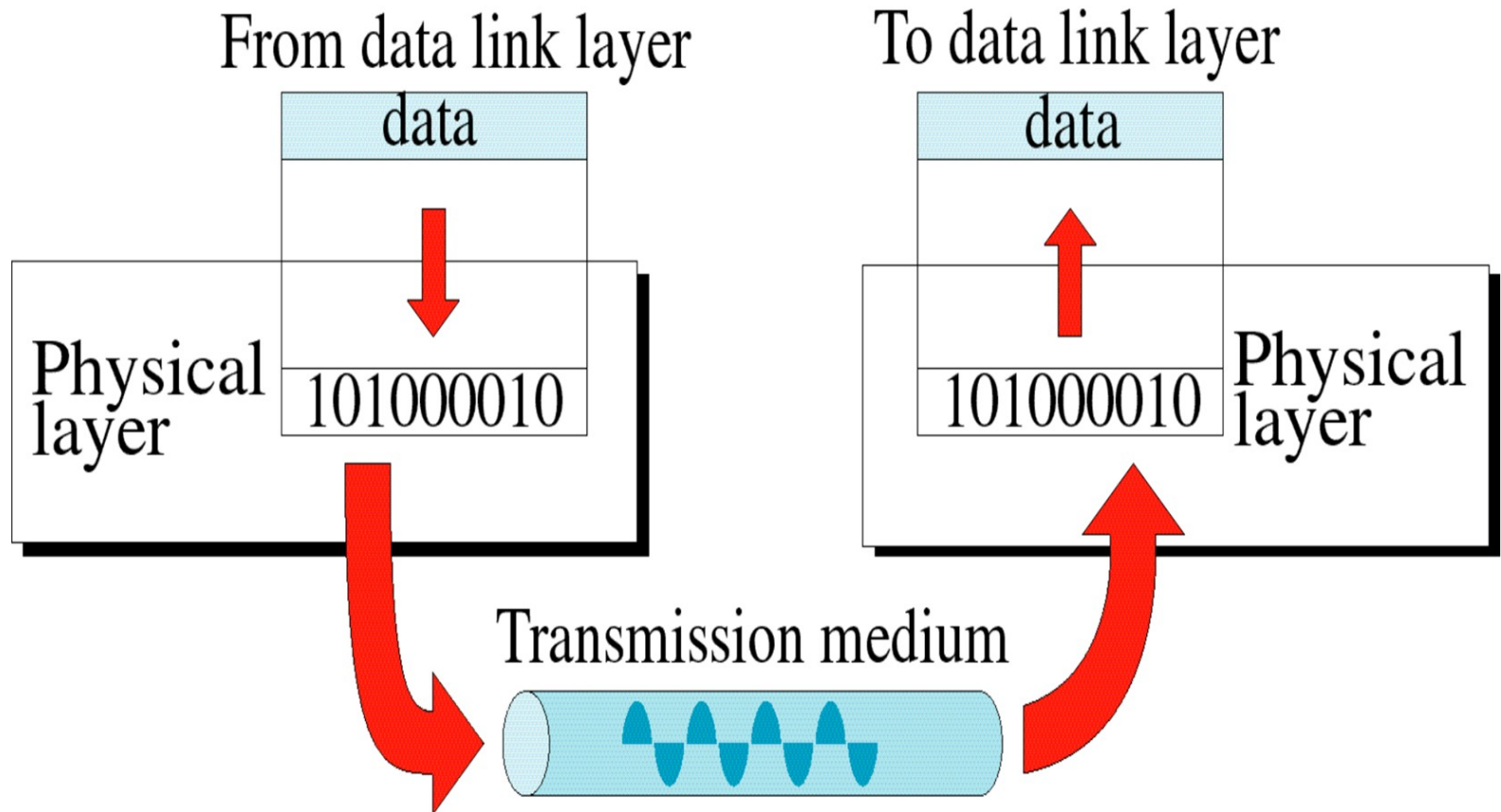
Physical Layer

Physical Layer

- It is responsible for the actual physical connection between the devices. Such physical connection may be made by using twisted pair cable.
- It is concerned with transmitting bits over a communication channel.
- the Physical layer is responsible for ultimate transmission of digital data bits from the Physical layer of the sending (source) device over network communications media to the Physical layer of the receiving (destination) device.

Physical Layer

- The Physical layer has two responsibilities: it sends bits and receives bits.
- The Physical layer communicates directly with the various types of actual communication media.
- Different kinds of media represent these bit values in different ways.



Cont..

The physical layer is concerned with :

- Physical characteristics of interfaces and medium:-
 - Representation of bits into signals
 - Data rate
- The Line configuration:-
 - Physical topology
 - Transmission mode

Signaling and encoding

- **Encoding** converts the stream of bits into a format recognizable by the next device in the network path.
- This ‘coding’ provides predictable patterns that can be recognized by the next device.
- The **signaling** method is how the bit values, “1” and “0” are represented on the physical medium.
- The method of signaling will vary based on the type of medium being used.

Bits

- Physical layer transports bits across the network media
- Accepts a complete frame from the Data Link Layer and encodes it as a series of signals that are transmitted to the local media
- This is the last step in the encapsulation process.
- The next device in the path to the destination receives the bits and re-encapsulates the frame, then decides what to do with it.

Ethernet

- Ethernet is the traditional technology for connecting devices in a wired local area network (LAN) or wide area network (WAN).
- It enables devices to communicate with each other via a protocol, which is a set of rules or common network language.
- Ethernet is used to connect devices in a network and is a popular form of network connection.

Point-to-Point Protocol

- It is a communication protocol of the data link layer that is used to transmit multiprotocol data between two directly connected (point-to-point) computers.
- one of the most common protocols for point-to-point access is the Point-to-Point Protocol (PPP).
- Today, millions of Internet users who need to connect their home computers to the server of an Internet service provider use PPP.

Point-to-Point Protocol

- The majority of these users have a traditional modem; they are connected to the Internet through a telephone line, which provides the services of the physical layer.
- But to control and manage the transfer of data, there is a need for a point-to-point protocol at the data link layer.
- PPP is by far the most common.

Cont..

PPP provides several services:

- PPP defines the format of the frame to be exchanged between devices.
- PPP defines how two devices can negotiate the establishment of the link and the exchange of data.
- PPP defines how network layer data are encapsulated in the data link frame.

Cont..

- PPP defines how two devices can authenticate each other.
- PPP provides multiple network layer services supporting a variety of network layer protocols.
- PPP provides connections over multiple links.
- PPP provides network address configuration. This is particularly useful when a home user needs a temporary network address to connect to the Internet.

Cont..

- On the other hand, to keep PPP simple, several services are missing:
- PPP does not provide flow control. A sender can send several frames one after another with no concern about overwhelming the receiver.
- PPP has a very simple mechanism for error control. A CRC field is used to detect errors.

Cont..

- If the frame is corrupted, it is silently discarded; the upper-layer protocol needs to take care of the problem.
- Lack of error control and sequence numbering may cause a packet to be received out of order.
- PPP does not provide a sophisticated addressing mechanism to handle frames in a multipoint configuration.

Types of network

- Networks are categorized based on the **geographical area they cover** and **their architecture**.
- Based on the geographical area they cover
 - PAN
 - LAN
 - MAN
 - WAN
- Based on their architecture
 - Peer-to-peer
 - Client-server

LAN (Local Area Networks) :

- LAN is a computer network widely used for local communication.
- LAN connects computers in a small area like a room, office, building or a campus spread up to a few kilometers.
- They are privately owned networks, to exchange information.

LAN Devices:

- Repeaters, Hubs, Bridges and switches

PAN (Personal Area Network)

- PAN is a network for interconnecting electronic devices within an individual person's workplace.
- It provides data transmission among devices such as computers, smartphones, tablets.

WAN(Wide Area Network)

- **WAN** is a network that connects computers over long distances like cities, countries, continents or world wide.
- **WAN** uses public, leased, or private communication links to spread over long distances. WAN uses telephone lines, satellite link and radio link to connect.
- Using INTERNET in public network is a common example of WAN.

WAN Devices

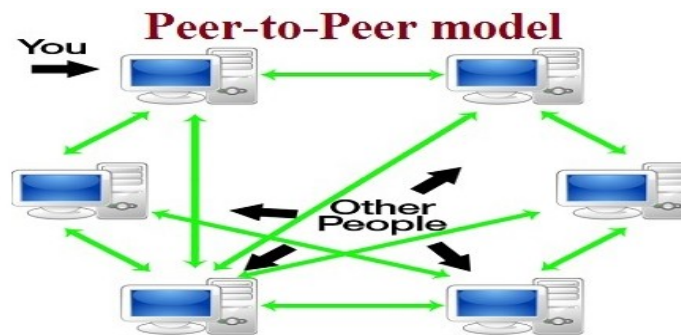
- Routers, Layer 3 Switches and Gateways

MAN (Metropolitan Area Network)

- MAN is a computer network spread over a city.
- The computers in a MAN are connected using cables.
- MAN connects several LAN spread over a city.
- Example- connection between branches

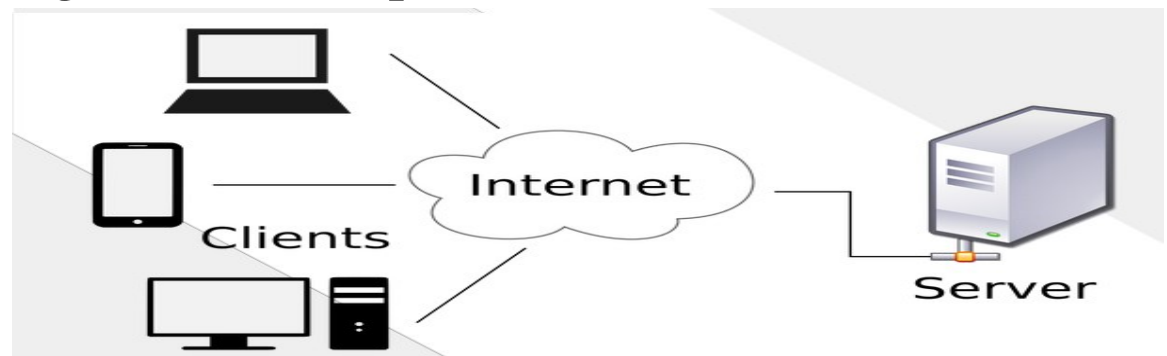
Peer-to-peer network

- Two or more PCs share files and access to devices such as printers without requiring a separate server computer or server software.



Client-server network

- Where client request for a service to a server and server replies according to client request.



Switching techniques

- These are techniques which are used to establish communication links between data, source and receiver in a communication network.
- There are three forms of switching techniques:-
 - Circuit switching
 - Packet switching
 - Message switching

Circuit Switching

- **In circuit switching**, a connection is established between two network nodes before they begin transmitting data.
- **Circuit switching** is a technique that directly connects the sender and the receiver in an unbroken path.
- With this type of switching technique, once a connection is established, a dedicated path exists between both ends until the connection is terminated.
- A circuit-switching network defines a static path from one point to another.
- As long as the two points are connected, all data traveling between those two points will take the same path.

Phases in Circuit Switching

- **Connection Establishment Phase**
- **Data Transfer Phase**
- **Connection Release Phase**

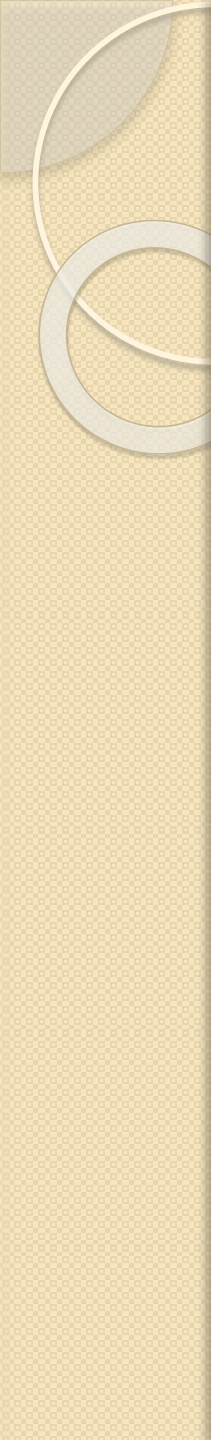
- E.g. Telephone line

Message switching

- In message switching there is no need to establish a dedicated path between two stations.
- It is also called as Store and forward technology.
- The information is stored and forwarded from the second device after a connection between that device and a third device on the path is established.
- In this technique a node receive a message and store the message until the free and appropriate route is found and if found then send it otherwise stored it.
- Message switching requires that each device in the data's path has sufficient memory and processing power to accept and store the information before passing it to the next node.

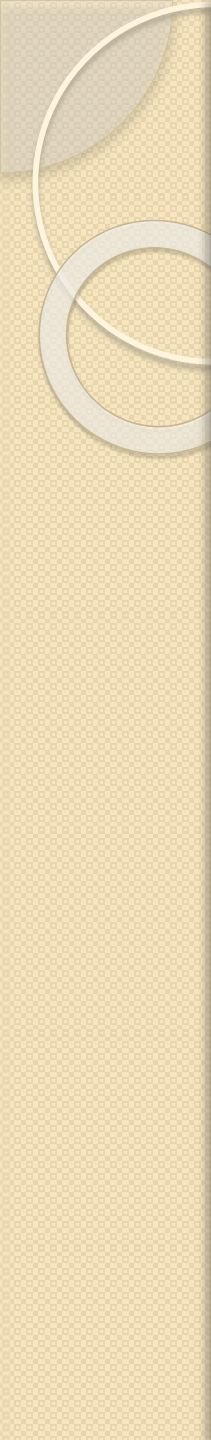
Packet Switching

- It can work in both connection oriented and connectionless.
- Packet switching breaks data into packets before they are transported.
- Packets can travel any path on the network to their destination, because each packet contains the destination address and sequencing information.



Physical Layer and its responsibilities

- Modulates the process of converting a signal from one form to another so that it can be physically transmitted over a communication channel
- Bit-by-bit delivery
- Line coding, which allows data to be sent by hardware devices that are optimized for digital communications that may have discrete timing on the transmission link
- Forward error correction/channel coding such as error correction code



End of chapter 5